

Module 08: Frontend Assets (Vite, Tailwind, Alpine.js)

Duration: 3-5 days | **Level:** Intermediate

Vite Integration

Laravel uses Vite for bundling CSS/JS:

```
@vite(['resources/css/app.css', 'resources/js/app.js'])
```

Commands

```
npm install
npm run dev      # Development with HMR
npm run build    # Production build
```

File Structure

```
resources/
|-- css/app.css      # Tailwind imports
|-- js/app.js        # Entry point
-- views/            # Blade templates
```

Tailwind CSS

Laravel 12 ships with Tailwind 4. Style in Blade:

```
<div class="bg-white p-6 rounded-lg shadow">
<h2 class="text-lg font-semibold text-gray-900">Tasks</h2>
</div>
```

Utility-first approach - no custom CSS files needed for most UI.

Alpine.js (Optional)

Lightweight JS for interactivity:

```
<div x-data="{ open: false }">
<button @click="open = !open">Toggle</button>
<div x-show="open">Content</div>
</div>
```

Livewire & Inertia (Advanced UI)

Tool	Use Case
Livewire	Full-stack components without writing API
Inertia.js	SPA feel with Vue/React + Laravel backend

```
composer require livewire/livewire
# or
composer require inertiajs/inertia-laravel
```

Production Assets

`npm run build`

Compiled files go to `public/build/`. Never run `npm run dev` in production.

Exercises

1. Run `npm run dev` and add a hover effect to task cards.
 2. Add Alpine.js dropdown for task status filter.
 3. Build assets and verify `@vite` works without dev server.
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Next Module

-> [Module 09: Validation & Form Requests](#)